

# Inhibitory effect of grapefruit juice and its bitter principal, naringenin, on CYP1A2 dependent metabolism of caffeine in man.

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1. The effects of grapefruit juice and naringenin on the activity of the human cytochrome P450 isoform CYP1A2 were evaluated using caffeine as a probe substrate. 2. In vitro naringin was a potent competitive inhibitor of caffeine 3-demethylation by human liver microsomes ( $K_i = 7\text{-}29\text{ }\mu\text{M}$ ). 3. In vivo grapefruit juice (1.2 l day<sup>-1</sup> containing 0.5 g l<sup>-1</sup> naringin, the glycone form of naringenin) decreased the oral clearance of caffeine by 23% (95% CI: 7%-30%) and prolonged its half-life by 31% (95% CI: 20%-44%) (n = 12). 4. We conclude that grapefruit juice and naringenin inhibit CYP1A2 activity in man. However, the small effect on caffeine clearance in vivo suggests that in general the ingestion of grapefruit juice should not cause clinically significant inhibition of the metabolism of other drugs that are substrates of CYP1A2.